



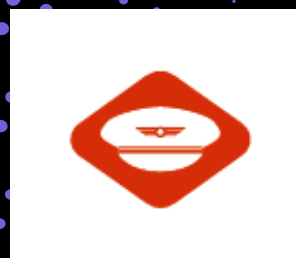
Network Segmentation



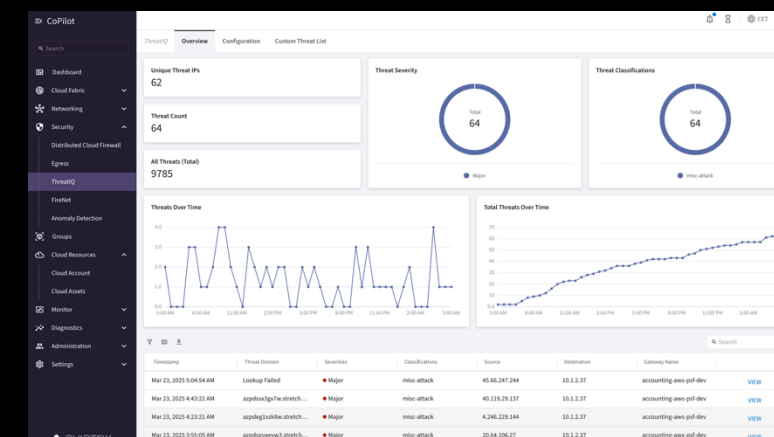
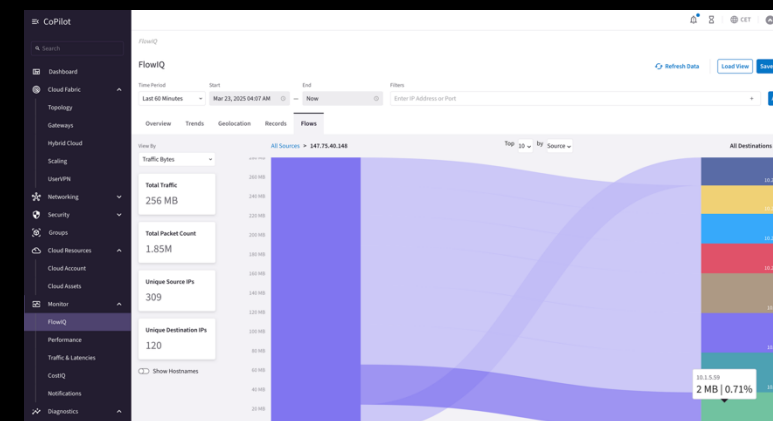
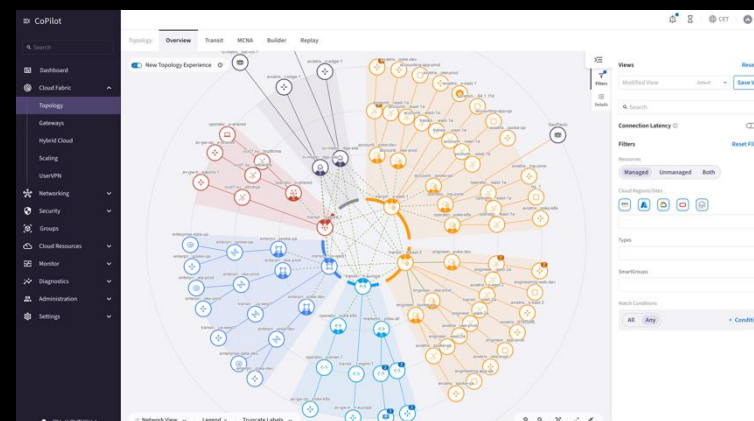
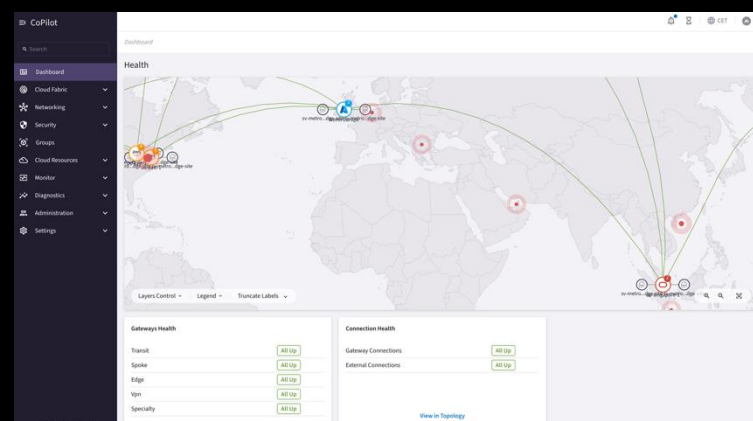
Segmentation

- **Main Purpose:** Enable ZTNA across multi-region and multi-cloud environments, including on-premises.
- Group VNets/VPCs/VCNs/Apps that share similar security policies.
- Define your own domains.
- Use Cases: Compliance, Governance, Audits.
- Network Segmentation is also referred to as **Macro-Segmentation**.
- A Network Domain can encompass one or more VPCs as a single logical container (i.e., **Routing Domain**).

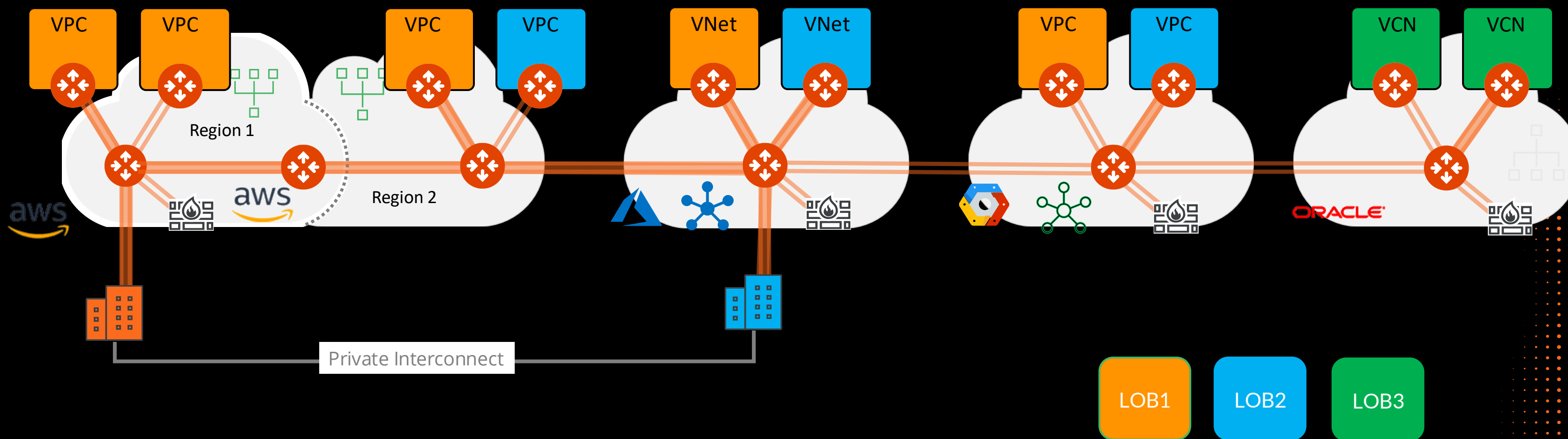
Hybrid-Cloud Network Segmentation



Aviatrix
CoPilot

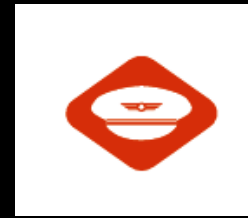


Network Granularity and Control



Hybrid-Cloud Network Segmentation

Aviatrix
CoPilot



Policy Based Network Segmentation

- Global
- Consistent / Repeatable
- Across accounts, subscriptions & projects

Cloud and Connection Agnostic

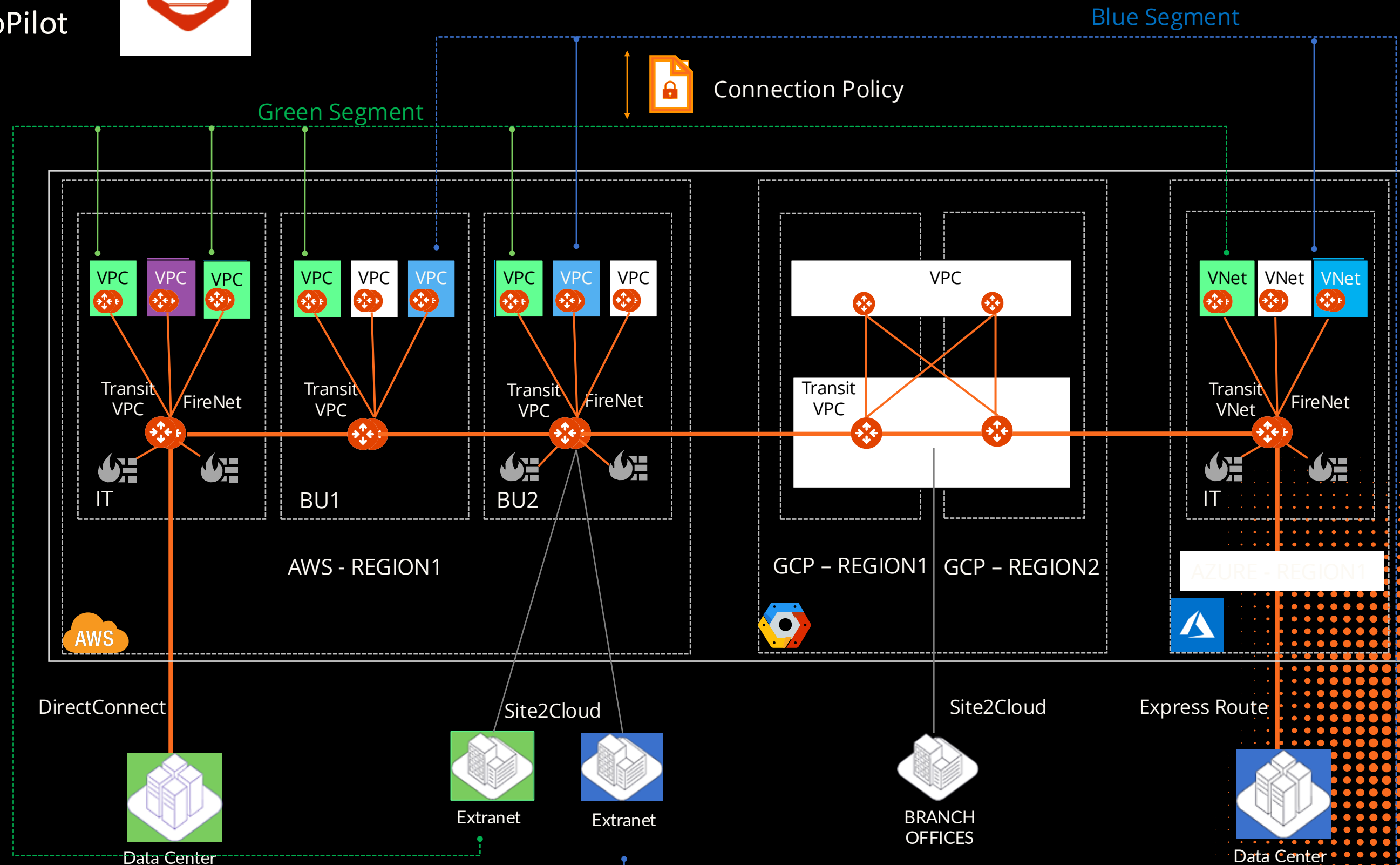
- Single cloud
- Intra-region or inter-region
- Multiple clouds

Edge/Access Segmentation

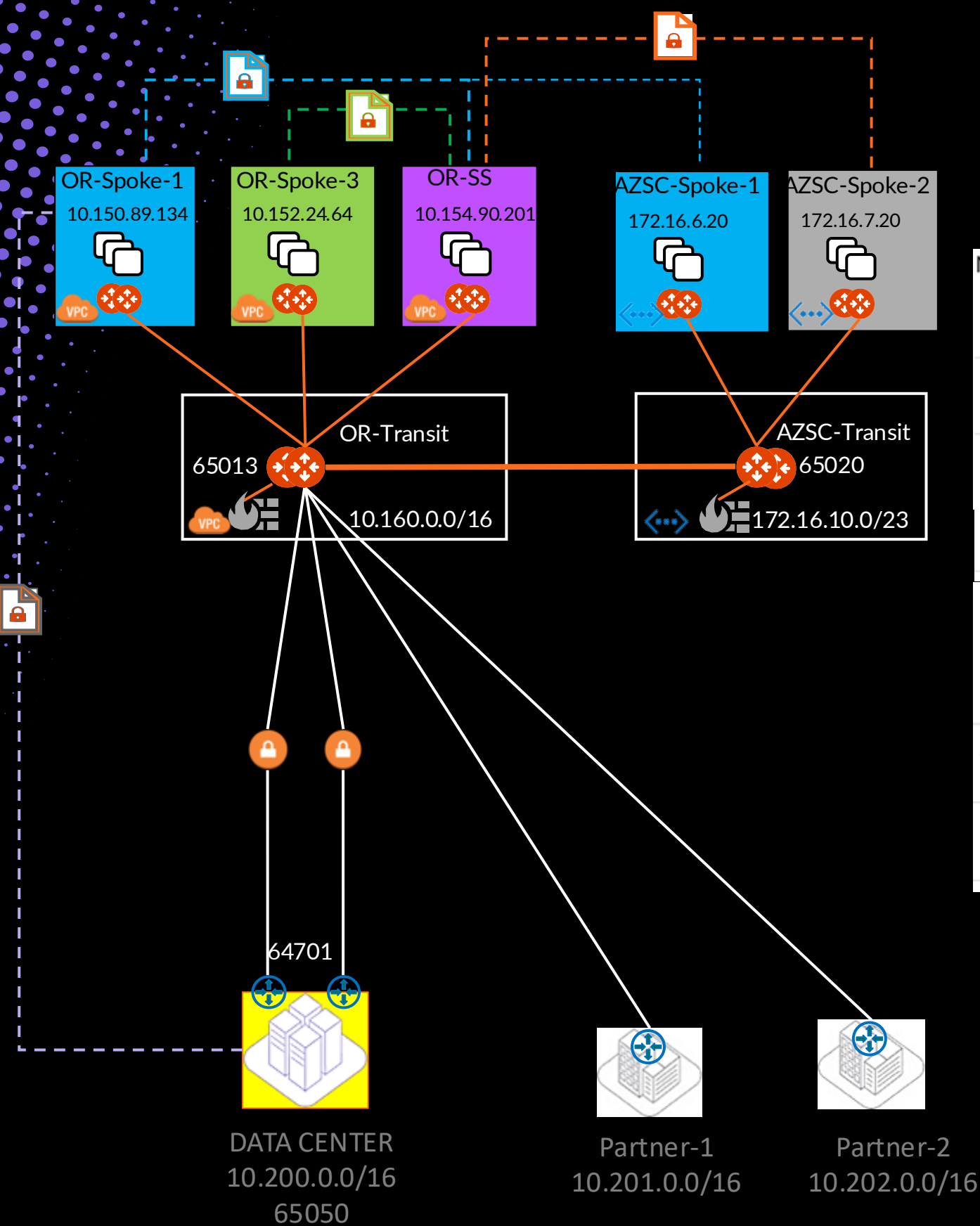
- On-Prem DCs
- Branches
- Extranets
- Cloud Peering

On-Demand Compliance/Governance

- Security Posture within minutes
- Aviatrix control plane realizes the intent
- Zero-Trust
- Flexible
- Automated



Hybrid-Cloud Network Segmentation



Name: AZSC-Spoke1-AGW

DESTINATION	VIA	DEV	NEXTHOP IP	NEXTHOP GATEWAY
default	172.16.6.65	eth0		
10.154.0.0/16		tun-AC100A44-0	172.16.10.68	AZSC-Transit-AGW
10.150.0.0/16		tun-AC100A44-0	172.16.10.68	AZSC-Transit-AGW
10.200.0.0/16		tun-AC100A44-0	172.16.10.68	AZSC-Transit-AGW
172.16.6.0/24	172.16.6.65	eth0		
172.16.6.64/26		eth0		
172.16.6.132		tun-3499E255-0	52.153.226.85	AZSC-Spoke1-AGW-hagw

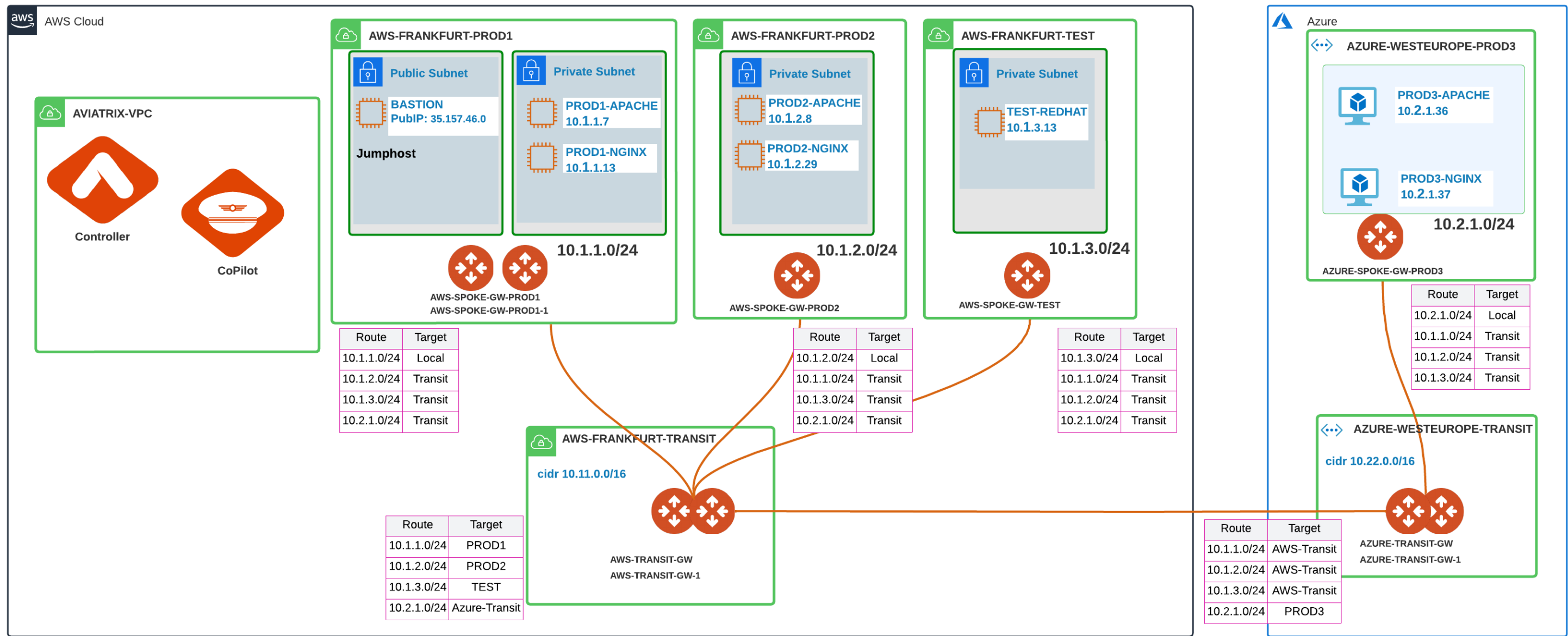
Purple

Remote-Blue

Yellow

Local-Blue

Enable Network Segmentation on the Transit Gateways



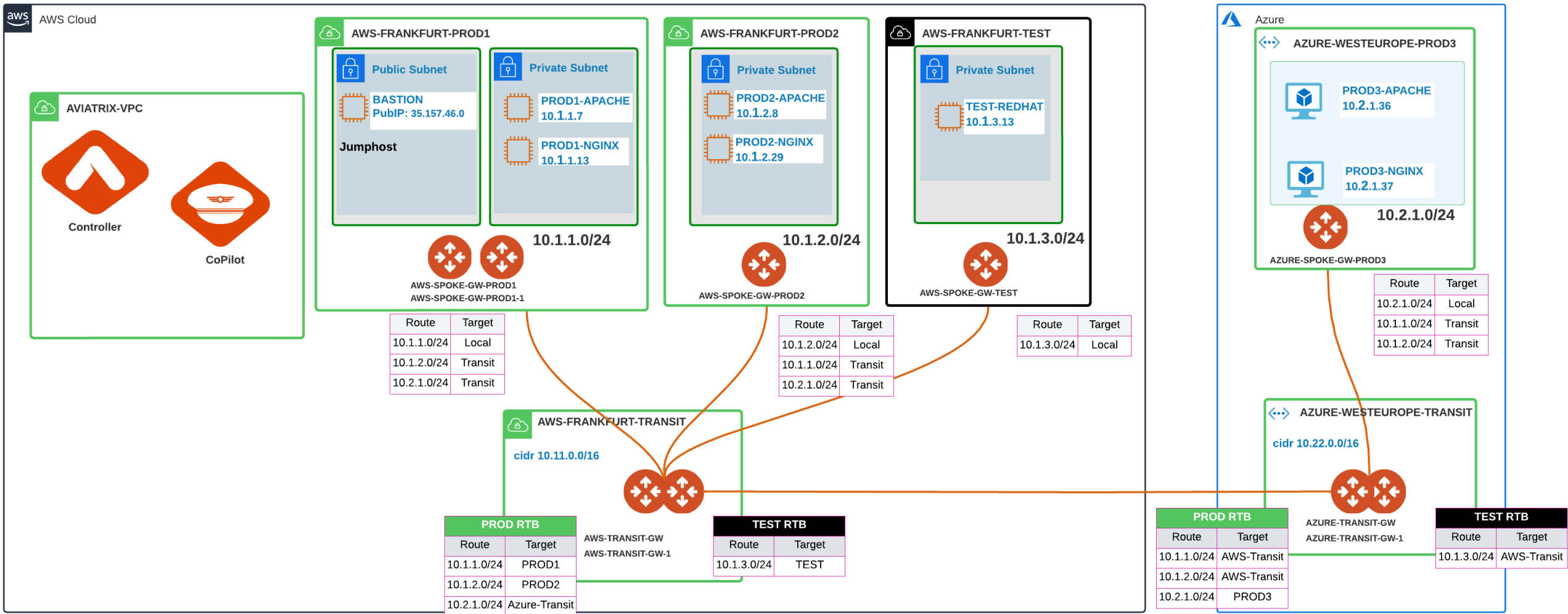
Configure Transit Gateways for Network Segmentation

Aviatrix transit gateways have to be enabled to support network segmentation on them.

Name	Cloud	Region	IP Address Space	
AWS-TRANSIT-GW	aws	eu-central-1	10.11.0.0/16	☒ Enabled
AZURE-TRANSIT-GW	arm	West Europe	10.22.0.0/16	☒ Enabled

Caveat: Select the Transit Gateways that will handle traffic for their associated members.

Creation of Network Domains and VPCs Association



Transit Gateway

- Multiple RTBs (per each Network Domain)
- Main RTB:
 - The main RTB will host the **Transit Routes** (i.e. the routes of the *backbone layer*) and the routes that belong to *Unmanaged Network Domains* (i.e. VPCs/Vnets not assigned to any Network Domains yet).

Spoke Gateway

- Single RTB (Main)

- Assign a Name to each Network Domain
- Associate the Spoke VPCs/Vnets and/or Site2Cloud Connections to the Network Domain

CAVEAT: You can create maximum **200** Network Domains per each Transit Gateway

Create Network Domain

Name *

PROD

Associations

AWS-FRANKFURT-PROD1 x AWS-FRANKFURT-PROD2 x

AZURE-WESTEUROPE-PROD3 x

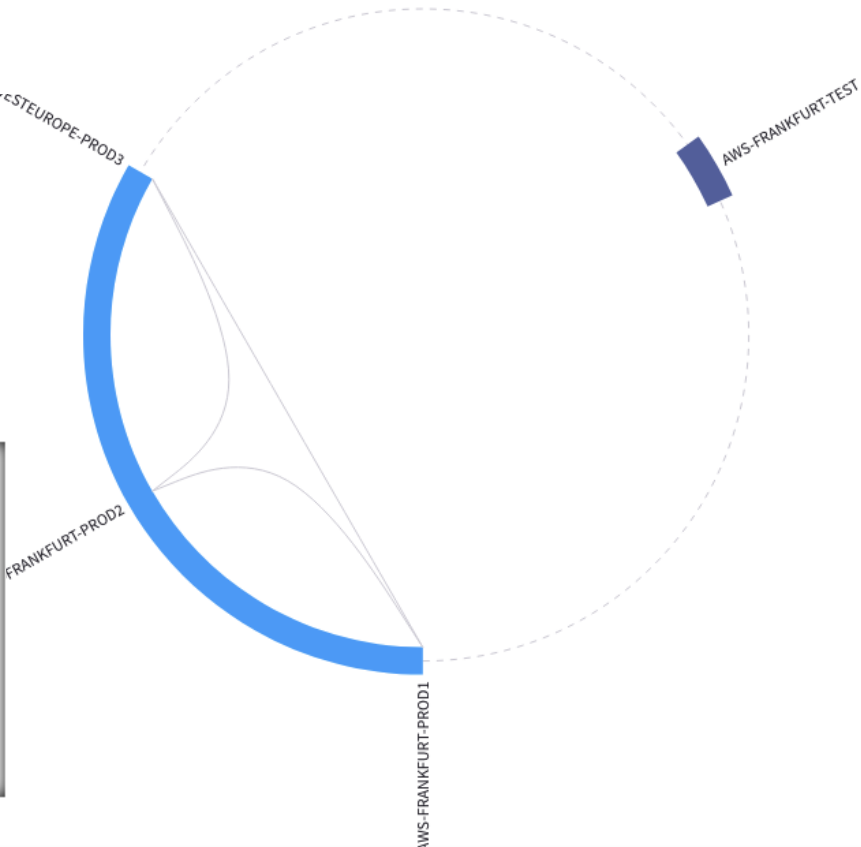
Create Network Domain

Name *

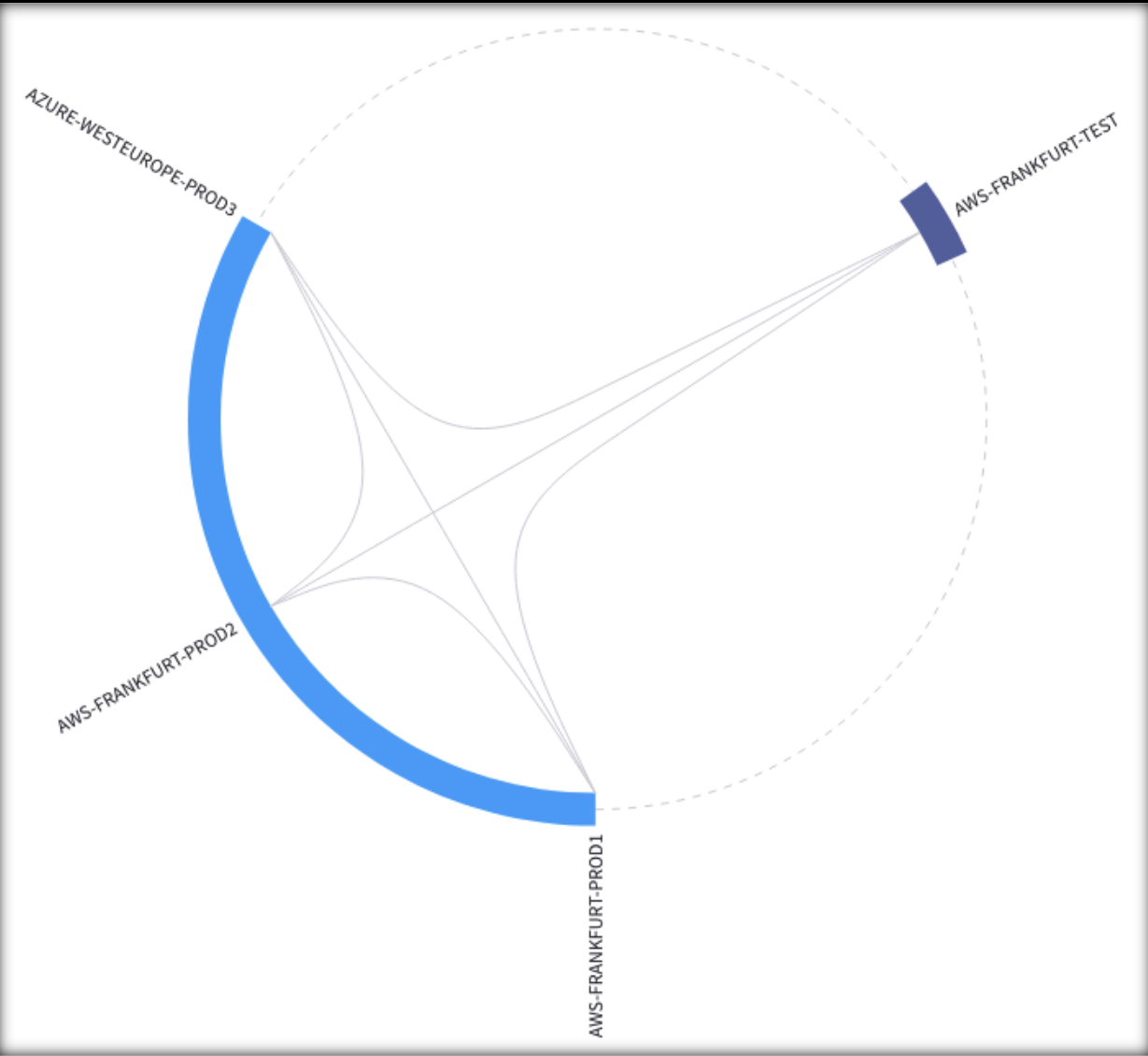
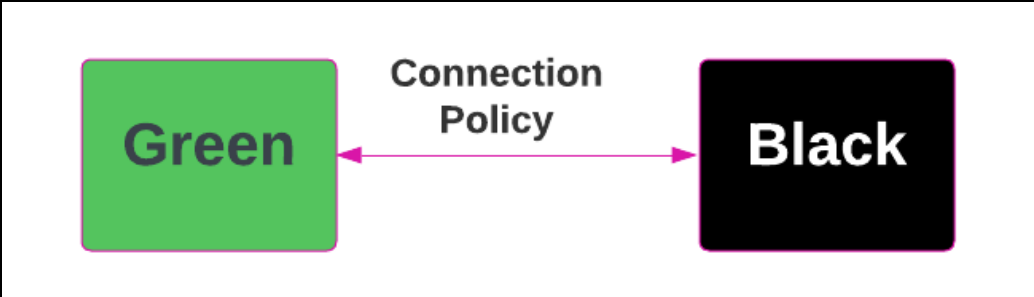
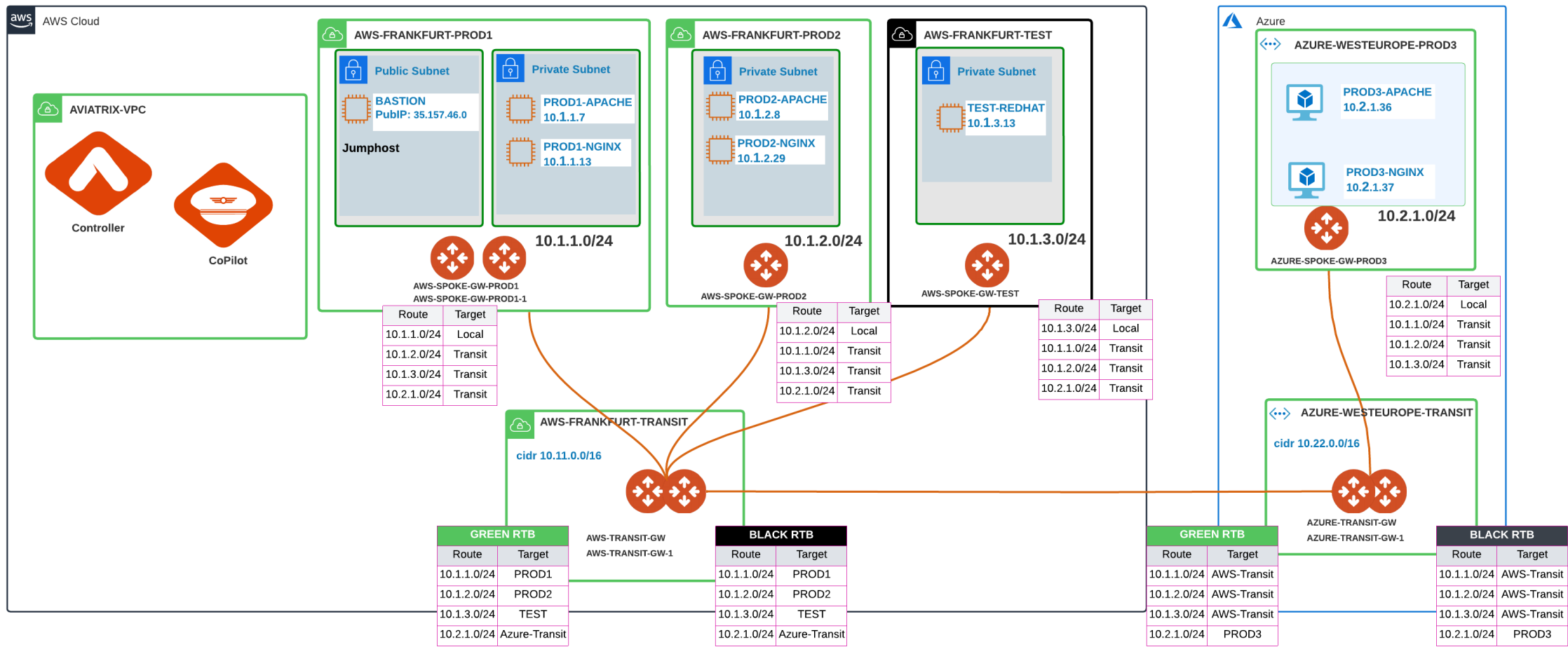
TEST

Associations

AWS-FRANKFURT-TEST x



Connection Policy



➤ Optionally enable the **Connection Policy**: the different routing tables will be merged (i.e., *VRF Lite Route Leaking*).

Edit Network Domain: PROD

Name *

PROD

Associations

AWS-FRANKFURT-PROD1

AWS-FRANKFURT-PROD2

AZURE-WESTEUROPE-PROD3

Connect to Network Domain

TEST

☒ TEST

Select All

Cancel

Save

Next: Lab 3 - Network Segmentation

